

MCCAMPBELL-PORTIER, TX, USA

WMO: 722369

Lat: 27.917N

Lon: 97.200W

Elev: 5

StdP: 101.26

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.3	3.1	-8.7	1.8	8.5	-5.6	2.4	10.5	12.0	15.6	11.1	16.3	4.0	0	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	35.2	25.3	34.4	25.6	33.9	25.7	27.6	31.8	27.3	31.5	26.9	31.1	5.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	22.1	29.2	26.2	21.7	29.1	26.0	21.3	29.1	87.1	31.7	85.6	31.7	84.3	31.1	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.9	8.9	DB	-2.1	37.7	2.7	1.3	-4.0	38.7	-5.5	39.4	-7.0	40.2	-9.0	41.1	
			WB	-3.7	27.9	2.4	0.6	-5.4	28.4	-6.8	28.7	-8.1	29.0	-9.9	29.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	13.9	16.5	19.3	22.9	26.0	28.7	29.9	30.1	28.3	24.7	19.0	15.5
	DBStd	6.80	5.00	5.10	4.62	3.15	3.01	2.78	1.27	1.32	2.18	3.87	5.10	4.99
	HDD10.0	49	21	9	5	0	0	0	0	0	0	0	4	10
	HDD18.3	466	153	85	44	5	1	2	0	0	0	5	54	118
	CDD10.0	4768	142	191	293	388	495	562	616	624	549	455	273	180
	CDD18.3	2144	15	33	75	143	237	314	358	366	299	202	73	29
	CDH23.3	23177	51	93	280	912	2343	3927	4658	4787	3487	2047	489	102
Wind	WSAvg	4.6	4.4	4.7	5.1	5.4	5.4	4.9	4.8	4.2	3.7	4.1	4.0	4.2
	PrecAvg	870	53	53	66	53	83	67	80	50	150	106	71	42
	PrecMax	1269	159	135	137	219	315	225	351	183	421	357	253	153
	PrecMin	462	0	2	2	5	0	0	0	1	33	0	0	8
	PrecStd	241	45	44	42	52	85	56	100	46	94	99	68	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	27.5	29.0	32.6	33.6	36.2	35.6	37.1	36.2	33.5	30.5	27.4
		MCWB	19.1	19.8	20.6	18.1	22.8	25.0	25.7	24.2	24.9	25.1	24.2	20.7
	2%	DB	24.9	25.9	27.2	29.9	32.2	34.0	34.5	35.2	34.3	32.4	28.8	26.0
		MCWB	19.5	20.9	21.6	21.7	24.4	25.7	25.9	25.5	25.3	25.6	23.4	21.3
	5%	DB	23.0	24.5	26.1	28.1	31.2	33.2	33.9	34.7	33.5	31.4	27.6	24.3
		MCWB	18.3	20.3	21.2	22.2	24.5	25.8	26.1	25.7	25.4	24.8	22.0	20.4
	10%	DB	22.0	22.8	25.0	27.4	30.2	32.6	33.2	33.9	32.7	30.3	26.2	22.8
		MCWB	18.4	19.5	20.5	22.1	24.4	25.7	26.1	25.8	25.4	23.9	21.6	20.1
	0.4%	WB	22.3	23.0	23.6	25.5	26.9	28.0	27.9	27.9	27.9	27.6	25.7	23.6
		MCDB	24.7	26.0	26.7	28.5	30.1	32.1	32.1	32.7	32.4	30.7	28.9	26.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.4	22.2	22.9	24.5	26.2	27.4	27.5	27.4	27.3	26.8	24.6	22.7
		MCDB	23.2	24.5	25.7	27.4	29.4	31.4	31.9	31.7	31.5	30.2	26.9	24.7
	5%	WB	20.6	21.5	22.4	23.9	25.8	26.9	27.2	26.9	26.9	26.3	23.6	21.9
		MCDB	22.1	23.3	24.8	26.7	29.0	31.0	31.5	31.3	31.0	29.7	26.0	23.2
	10%	WB	19.2	20.8	21.7	23.4	25.4	26.6	26.9	26.7	26.5	25.7	22.9	21.0
		MCDB	20.9	22.1	23.9	26.0	28.7	30.7	31.2	31.1	30.6	29.1	25.4	22.5
Mean Daily Temperature Range	5% DB	MDBR	10.4	9.3	8.6	8.3	7.5	7.3	7.1	8.2	8.5	9.6	10.0	9.9
		MCDBR	11.5	9.2	7.7	8.7	7.8	7.5	7.8	9.1	9.2	8.6	9.7	9.7
		MCWBR	7.2	5.4	3.9	4.2	2.7	2.2	2.3	2.6	3.1	3.6	5.4	6.5
	5% WB	MCDBR	8.7	7.7	6.9	6.3	5.9	6.5	6.7	7.5	7.3	7.4	8.4	8.6
		MCWBR	6.8	5.5	4.0	3.4	2.4	2.3	2.2	2.3	2.8	3.9	5.5	6.8
Clear-Sky Solar Irradiance	taub		0.361	0.367	0.387	0.429	0.471	0.448	0.465	0.449	0.439	0.393	0.378	0.349
	taud		2.476	2.461	2.402	2.291	2.205	2.333	2.303	2.346	2.337	2.462	2.453	2.525
	Ebn at Noon		881	904	904	871	830	845	829	842	842	866	853	878
	Edh at Noon		95	104	117	134	146	128	132	125	123	103	97	87
All-Sky Solar Radiation	RadAvg		2.98	3.53	4.53	5.39	6.06	6.68	6.58	6.29	5.21	4.53	3.40	2.69
	RadStd		0.33	0.58	0.47	0.51	0.43	0.35	0.56	0.35	0.45	0.47	0.36	0.36

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+81

Nomenclature: See separate page